

# Billa Abhinav

B.Tech in Mathematics and Computing



| ACADEMIC PROFILE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |                              |               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------------------------------|---------------|
| B.Tech - B.Tech in Mathematics and Computing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 7.7  | Indian Institute of Science  | 2028          |
| Class XII(Board of Intermediate Education Andhra Pradesh (BIEAP))                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 98.1 | SRI CHAITANYA JUNIOR COLLEGE | 2024          |
| Class X(SSC)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 97.0 | SRI CHAITANYA SCHOOL         | 2022          |
| COURSES                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |      |                              |               |
| Data Structures and Algorithms (UG)<br>Probability and Statistics (UG)<br>Introduction to Artificial Intelligence and Machine Learning (UG)<br>Automata and Computability (UG)<br>Discrete Mathematics (UG)<br>Analysis and Linear Algebra I (UG)<br>Analysis and Linear Algebra II (UG)<br>Numerical Methods (UG)<br>Algebraic Structures (UG)                                                                                                                                                                                                                                                                                                                                                                                           |      |                              |               |
| PROJECTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |      |                              |               |
| Variational Autoencoders                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |      | Technical Lead               | Mar'26-Apr'26 |
| <ul style="list-style-type: none"><li>Implemented and evaluated continuous (Standard VAE), modified-objective (InfoVAE), and discrete (VQ-VAE) generative frameworks on CIFAR-10 and CelebA benchmark datasets.</li><li>Investigated core training dynamics, focusing on posterior collapse and latent space utilization by tracking empirical KL divergence trends and codebook utilization constraints.</li><li>Conducted linear latent space interpolations and quantified output quality using FID and LPIPS metrics to analyze the trade-off between representation continuity and semantic sharpness.</li></ul>                                                                                                                     |      |                              |               |
| Drift Monitoring API – ML Model Monitoring Platform                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |      | Developer                    | May'26-Jun'26 |
| <ul style="list-style-type: none"><li>Built a drift detection platform with a FastAPI backend, Streamlit dashboard, and SQLite persistence layer to monitor and flag statistical drift in structured data pipelines.</li><li>Engineered a modular detection pipeline supporting Kolmogorov-Smirnov (continuous) and Population Stability Index (categorical) methods, with a generalized column profiler and human-in-the-loop schema verification to ensure robustness across diverse structured file formats.</li><li>Built a validation suite to benchmark detection accuracy across batch sizes on the NYC Citi Bike dataset, identified 15,000 rows as the batch size yielding strongest F1 performance for that workload.</li></ul> |      |                              |               |
| ACHIEVEMENTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |      |                              |               |
| <ul style="list-style-type: none"><li>Secured All India Rank (AIR) 3148 in JEE Advanced</li><li>Secured All India Rank (AIR) 1260 in JEE Mains</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |      |                              |               |
| POSITION OF RESPONSIBILITY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |      |                              |               |
| <ul style="list-style-type: none"><li>Core Coordinator of the Merchandise Team for Pravega Fest</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |      |                              |               |
| LANGUAGES                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |      | English, Telugu, Hindi       |               |